Renxiang Chu

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EDUCATION

Peking University **Beijing, China**

- Bachelor of Medicine in Basic Medical Sciences *Expected Jul. 2027*
  Dual Degree in Economics

Core Courses:

Medical Molecular and Cell Genetics, Organic Chemistry, Nervous System, Introduction to Computing (B), Principles of Economics, Probability and Statistics (B), Macroeconomics and Health

RESEARCH EXPERIENCE

PKU Basic Medical Sciences Innovative Talent Program **Position:** Research Trainee
Supervisor: *Tingting Li* *Sept. 2023 - Aug. 2024 | Beijing*

- Studied the computational characterization of biomolecular condensates in a bioinformatics research group.
- Developed R/Python pipelines to quantify condensate composition and phase-separation dynamics, integrating literature and experimental evidence to examine macromolecular assembly stability.

Peking University Institute for Advanced Clinical Medicine **Position:** Research Assistant
Supervisor: *Meng Lu* *Sept. 2024 - June 2027 | Beijing*

PROJECT: Retinal Aging and Biological Age Modeling

- Built a VisionFM-based retinal aging clock on 112,019 fundus images, achieving a mean absolute error of 2.49 years.
- Developed teacher-student residual learning to separate physiological aging signals from pathological variation; co-first-authored a medRxiv preprint (DOI: 10.64898/2026.02.13.26346265).

PROJECT: SciEngram - AI Co-Scientist Memory

- Designed SciEngram, a traceable cognitive-memory architecture and benchmark for long-term scientific reasoning across papers.
- Evaluated evidence attribution, cross-paper reasoning, conflict detection, and belief updating against long-context, RAG, and general-purpose memory baselines; co-first-authored a submission to ICLR 2027.

Peking University School of Basic Medical Sciences **Position:** Undergraduate Researcher (Part-time)
Supervisor: *Guangxi Wang* *Mar. 2026 - June 2027 | Beijing*

PROJECT: AI Co-Scientist for Cancer Biomarker Discovery

- Developed machine-learning and deep-learning multi-omics models and an automated workflow for cancer biomarker discovery.

PROJECT: HistoSeek - AI-Assisted Histology Education | Co-supervisor: *Xiaofan Wei*

- Designed HistoSeek, a teacher-governed platform for visual-localization practice and personalized feedback, and evaluated it in an undergraduate histology course.
- Co-first-authored the manuscript, currently under external review at npj Digital Medicine.

Northeastern University Khoury College of Computer Science **Position:** Intern (Remote)
Supervisor: *Wengong Jin* *Mar. 2026 - Dec. 2026 | Boston*

PROJECT: World Models for Clinical Trial Simulation

- Developed an action-conditioned world model using MIMIC and a private heart-failure dataset to simulate patient trajectories under candidate treatments for offline clinical-trial evaluation.
- Proposed RCT-based evaluation of world-model counterfactual reasoning and screened suitable trials; contributed causal tracing and model evaluation; co-authored an accepted WMHS Workshop paper at NeurIPS 2026.

PUBLICATIONS

Chu, R.#, Sun, A.#, Qu, J., & Lu, M. (2026). ***Disentangling physiological heterogeneity in retinal aging using a deep learning-based biological age framework*** (https://doi.org/10.64898/2026.02.13.26346265). ***medRxiv*** (Preprint)

Chu, R.#, Sun, C.#, Zhang, Y.#, Sun, Z.#, Chen, Y., Shang, Y., Jiang, Y., & Lu, M. (2026). ***SciEngram: Evidence-grounded scientific memory for cumulative research reasoning***. Manuscript submitted to ***ICLR 2027***

Chu, R.#, Chi, X.#, Li, Z., Zhang, X., Xu, J., Wang, G., & Wei, X. (2026). ***HistoSeek links histology image recognition to teacher-authored formative guidance***. Manuscript under review at ***npj Digital Medicine***

Luo, T., **Chu, R.**, Shi, Z., Li, X., Li, Q., & Jin, W. (2026). ***One Patient, Two Futures: World Models for Clinical Trial Simulation***. Accepted at the ***WMHS Workshop at NeurIPS 2026***.

Equal contribution

INDUSTRY EXPERIENCE

Huagai Capital **Position:** Venture Capital Intern
Medical Devices & Technology Team *Sept. 2026 - Jan. 2027 | Beijing*

- Conducted investment-oriented industry research on brain-computer interfaces and AI for biology (AI4Bio), including AI-enabled molecular and drug discovery, virtual-cell models, laboratory automation, and DNA foundation models.
- Conducted on-site diligence of early-stage startups and interviewed founders to assess technical differentiation, product roadmaps, and commercialization potential.

LEADERSHIP & SERVICE

Team Leader, Peking University School of Basic Medical Sciences Social Practice Group

- Organized the Peking University School of Basic Medical Sciences social practice delegation to Suzhou under the theme "Exploring Development Achievements, Fostering the Spirit of Innovation," and served as team leader.
- Visited Soochow University and Kunshan Biomedical Industrial Park and met with Synermore Biologic Co., Ltd.

Bass-Baritone Vocalist, Peking University Health Science Center A Cappella Ensemble

- Performed bass and baritone parts in multiple ensemble performances.

SKILLS

- **Skills:** *MS Office (Proficient); Python/R (Working Knowledge)*
- **Tools & Platforms:** *Linux, AWS, Conda, Jupyter, NumPy, Pandas, LaTeX, Git, Markdown*
- **Languages:** *English (CET-6: 607; TOEFL: 106); Mandarin Chinese (Native)*